

Sr. No.	Description	UNIT	Total Qty
E	CHILLED WATER PIPING & VALVES:		
1	Supply, installation, testing & commissioning of thermally insulated MS Class C (heavy) chilled water piping as per IS1239 / IS 3589 (8mm pipe thickness) complete with bends, elbows, reducers all fittings (provision to install sensor of BMS) & supports as required as per enclosed specification. anti microbial insulation for chilled water piping. The insulation shall be closed cell nitrile rubber insulation class O density 48kg/m3, covered with canvass & vapour barrier shall be 18 oz. teflon fiberglass cloth/covered with canvass & vapour barrier Foster compound / Lag coating+cora cloth. Insulation shall be fixed with adhesive as per specifications. 32 mm thick insulation for Pipe size upto 125mm dia. and 50 mm thick for pipe size above 125mm diameter. Multi layer insulation shall be with staggered seams. Final finished colour of coating will be as per approved shade. (Inside Piping) Insulation color should be as per following : - Internal Piping - CHWS: Blue - CHWR: Gray - External Piping - CHWS & CHWR: Black w/ Aluminum Cladding complies for closed cell NBR with fiberglass cloth of 225 gsm made from E glass and vapor control layer of 165 micron		COMPLY
1.1	500 mm dia	Rmt.	0
1.1	400 mm dia	Rmt.	0
1.2	350 mm dia	Rmt.	0
1.3	300 mm dia	Rmt.	1540
1.4	250 mm dia	Rmt.	365
1.5	200 mm dia	Rmt.	520
1.6	150 mm dia	Rmt.	1288
1.7	125 mm dia	Rmt.	1114
1.8	100 mm dia	Rmt.	4506
1.9	80 mm dia	Rmt.	2777
1.10	65 mm dia	Rmt.	2534
1.11	50 mm dia	Rmt.	277
1.12	40 mm dia	Rmt.	9
1.13	32 mm dia	Rmt.	660
1.14	25 mm dia	Rmt.	1390
2	Supply, installation, testing & commissioning of thermally insulated MS Class C (heavy) chilled water piping as per IS1239 / IS 3589 (8mm pipe thickness) complete with bends, elbows, reducers all fittings (provision to install sensor of BMS) & supports as required as per enclosed specification. The insulation shall be PUF Insulation with 32 kg/m3 density & K value shall be minimum 0.025 W/m K. 50mm thick for Pipe size upto 200mm dia. and 75mm thick for pipe size above 200mm dia. should be covered with vapour barrier shall be 18 oz. teflon fiberglass cloth/Insulation with Lag Coating as Vapour Barrier and 22 Gauge thick AL Cladding. (Shaft & terrace piping) Insulation color should be as per following : - Internal Piping - CHWS: Blue - CHWR: Gray - External Piping - CHWS & CHWR: Black w/ Aluminum Cladding Remark: PUF insulation generally complies on material, thermal performance, and thickness; with fiberglass cloth of 225 gsm made from E glass and vapor control layer of 165 micron		COMPLY

Sr. No.	Description	UNIT	Total Qty
2.1	500 mm dia	Rmt.	1044
2.1	400 mm dia	Rmt.	0
2.2	350 mm dia	Rmt.	0
2.3	300 mm dia	Rmt.	146
2.4	250 mm dia	Rmt.	0
2.5	200 mm dia	Rmt.	0
2.6	150 mm dia	Rmt.	0
2.7	125 mm dia	Rmt.	0
2.8	100 mm dia	Rmt.	0
2.9	80 mm dia	Rmt.	0
2.10	65 mm dia	Rmt.	0
2.11	50 mm dia	Rmt.	0
2.12	40 mm dia	Rmt.	0
2.13	32 mm dia	Rmt.	0
2.14	25 mm dia	Rmt.	0
3	Supply, installation, testing & commissioning of thermally insulated MS Class C (heavy) chilled water piping as per IS1239 / IS 3589 (8mm pipe thickness) complete with bends, elbows, reducers all fittings (provision to install sensor of BMS) & supports as required as per enclosed specification for terrace and shaft piping. The insulation shall be Elastomeric insulation with integrated polymeric cladding (Armaclad dura/equivalent) Insulation shall be fixed with adhesive as per specifications. 32 mm thick for Pipe size upto 125mm dia. and 50 mm thick for pipe size above 125mm dia. Final finished colour of coating will be as per approved shade. Multi layer insulation shall be with staggered seams. (Inside Piping) Insulation color should be as per following : - Internal Piping - CHWS: Blue - CHWR: Gray - External Piping - CHWS & CHWR: Black w/ Aluminum Cladding complies for class O NBR with duraclad		
3.1	400 mm dia	Rmt.	0
3.2	350 mm dia	Rmt.	0
3.3	300 mm dia	Rmt.	0
3.4	250 mm dia	Rmt.	0
3.5	200 mm dia	Rmt.	0
3.6	150 mm dia	Rmt.	0
3.7	125 mm dia	Rmt.	0
3.8	100 mm dia	Rmt.	0
3.9	80 mm dia	Rmt.	0
3.10	65 mm dia	Rmt.	0
3.11	50 mm dia	Rmt.	0
3.12	40 mm dia	Rmt.	0
3.13	32 mm dia	Rmt.	0